

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B.TECH DEGREE EXAMINATION (Sample paper in KTU format)

Course Code: CST201 · Course Name: DATA STRUCTURES

Max. Marks: 23

Duration: 1 Hour

PART A

Answer all questions. Each question carries 3 marks.

- 1** Define time complexity. What is Big-O notation? **(3)**
- 2** Differentiate between a stack and a queue. **(3)**
- 3** Convert the infix expression $A + B * C$ into postfix form. **(3)**

PART B

Answer any one full question. Each question carries 14 marks.

- 4** a) Write an algorithm to insert a node at the beginning of a singly linked list. **(7)**
b) Explain a circular queue with a neat diagram. **(7)**

OR

- 5** a) Explain binary search with a suitable example. **(7)**
b) Write the bubble sort algorithm and find its time complexity. **(7)**

Sample question paper in the KTU format, made for testing Mark Calculator. It is not an actual KTU examination paper.